

Fig. 2: An actual-size, single-side PCB for multi-tone configurable alarm

T3 increases. Once it reaches beyond the threshold established with VR2, transistor T3 conducts and reset pin 12 of IC1 goes 'low.'

The make of NTC1 is not critical but VR2 should have an appropriate value depending on the selected NTC. The counter in IC1 starts advancing with rise in temperature and square-wave signals start appearing on the outputs. Through push-on switches S1-S10 you may select the frequencies required in the audio alarm signal. The produced composite signal available at the base

of transistor T1 is amplified to drive loudspeaker LS1. The mixed multi-tone signal produced is generally difficult to ignore.

Though 5V power supply is recommended here, the circuit works with 2V to 6V. Also, CD4060 may be replaced with 74HCT4060 or 74HC4060 for IC1. LED1 indicates the presence of power supply, and resistor R16 is for current limiting.

Construction and testing

An actual-size, single-side PCB for the alarm is shown in Fig. 2 and its

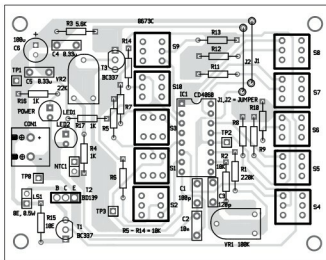


Fig. 3: Component layout for the PCB

component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable plastic case. Fix 2-pin terminal block connector CON1 on the PCB for power supply. Connect NTC1 through external wires for temperature sensing. Fix potentiometers VR1 and VR2 on the front panel for frequency control and threshold level settings. ●

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